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SerializableSimpy: Parallel and Serializable Discrete-Event Simulation in Python

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Summary

SerializableSimpy is a Python framework for building discrete-event simulations (DES). DES modeling and analysis are useful in time-sensitive applications, such as manufacturing, logistics, and distributed systems. The framework provides core DES building blocks, including logical processes (LPs) and event causality. It also includes components for state and synchronization, such as stores, resources, and priority queues. Users focuse on business logic, not the event-processing engine. After modeling the system, users can start, pause, and resume.

SerializableSimpy is inspired by SimPy (Zinoviev, 2024) but replaces its generator-based execution engine with a serializable design that enables checkpoint/restart and parallel or distributed runs (Fujimoto, 2017). It preserves familiar modeling abstractions while focusing on reproducibility and scaling. Unlike SimPy, which relies on non-serializable generator code (Vassalotti, 2009), SerializableSimpy implements context switches between LPs using standard Python function calls, with an event queue of Python callables. In other words, code that would use *yield* in SimPy is replaced with functions that finish using *return*.

This design enables:

- **Effortless checkpointing and full recovery** of simulation state, including LPs and event queues, using standard serialization tools such as pickle (Python Software Foundation, 2023b). It requires no serialization logic from users.
- **Parallel Discrete-Event Simulation** (Fujimoto, 2017) using `multiprocessing` (Python Software Foundation, 2023a) or `mpi4py` (Dalcin et al., 2011), with support for inter-process event exchange through shared memory or message passing.

SerializableSimpy reuses familiar SimPy abstractions where possible, while offering a fundamentally different execution engine. It is not a drop-in replacement, it is designed for users who require reproducibility, parallel performance scaling, or integration into networked systems.

The package includes tutorials, reproduces selected official SimPy examples, and provides performance comparisons with other DES frameworks.

Statement of Need

SimPy is widely adopted in both academia and industry due to its simplicity and expressiveness. However, its reliance on Python generators makes it unsuitable for simulations requiring state checkpointing, multiprocessing, or execution across machines in a computer network. This limits its use in modern digital twin workflows, especially in large-scale applications.

SerializableSimpy addresses this need by providing a fully serializable DES framework that offers API similar to SimPy but avoids generators internally. By replacing generator-based flow control with callback-based logic, SerializableSimpy enables features such as:

- Running simulations in parallel with inter-process communication.
- Events portability across networked systems.
- State checkpointing of the simulation environment (and its internal event queue) using serialization.

State of the PDES Field

Over the years, a number of open-source parallel discrete-event simulation (PDES) frameworks have been developed to efficiently model systems and processes involving very large numbers of events. While they differ in scope and design, they share a common goal: to provide software developers and scientists with building blocks for constructing efficient simulators. It is important to note that these frameworks are not simulators themselves, but rather foundational components for composing and processing events.

The table below provides an overview of selected general-purpose PDES tools, focusing on their implementation languages, most recent known activity, and the core parallelization or decomposition algorithms they employ for event processing. This comparison is intended to contextualize SerializableSimpy's contributions within the broader landscape of discrete-event simulation for large-scale systems.

The selected tools PDES software are:

- Root-sim (Pellegrini, 2021; Vitali et al., 2012)
- ROSS (Carothers et al., 2022; Gonsiorowski, 2016)
- Simian (Santhi et al., 2015; Seshadri & others, 2023)
- SimX (Los Alamos National Security, 2014; Thulasidasan et al., 2014)
- SST-core (SST Team, 2024)
- xSim Poshtkohi (2024)
- Warped2 (W. Team, 2020)
- Simulus (Liu, 2020; S. Team, 2019)
- PowerDEVS (Bergero & Kofman, 2011; Power-devs Team, n.d.)
- Simpy (SimPy team, n.d.; Zinoviev, 2024)
- SerializableSimpy (Pochelu, 2025)

Although SimPy is not primarily intended for parallel simulation, it is included in the table because of its popularity, expressiveness, and simplicity. It remains a widely used framework for rapid prototyping and is often the framework of choice for computationally less demanding simulations.

Name	Language	Parallel Algorithms
ROOT-sim	C	Optimistic, anti-messages, DyMeLoR memory manager (Pellegrini et al., 2009; Toccaceli & Quaglia, 2008)
ROSS	C	Optimistic & conservative, LZA compression
Simian	Lua, Python, JS	Conservative, some optimistic support, GPU, Greenlet (Greenlet Team, 2024)
SimX	C++, Python	Conservative, Greenlet (Greenlet Team, 2024)
SST-core	C++	Conservative, threading, graph-based LP organization
xSim	C, Fortran	Optimistic w/o rollback; METIS, topological sort, Tarjan (Tarjan, 1972)
Warped2	C++	Optimistic, Ladder Queue (Tang et al., 2005), METIS
Simulus	Python	YAWNS sync protocol (Nicol, 1993), Greenlet (Greenlet Team, 2024)
PowerDEVS	C++	Conservative, hierarchical model construction, real-time target
Simpy	Python	N/A

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Name	Language	Parallel Algorithms
SerializableSimpy	Python	Conservative

Experimental highlights (token-ring benchmark):

- **Familiar, rich API:** 19 classes at publication time, inspired by SimPy's 35 classes. Other frameworks include Simian (7 classes) and Simulus (15 classes).
- **Compact event semantics for scale:** ~1M events in the benchmark vs ~4M for SimPy, ~1M for Simian, ~2M for Simulus — fewer queue operations overall.
- **Fastest runtime:** best initialization and main-loop times, driven by fewer event operations and efficient heapq-based priority queues.
- **Parallel-ready:** supports multiprocessing and MPI backends.

Detailed scripts and results: https://gitlab.com/uniluxembourg/hpc/research/cadom/serializable-simpy/-/blob/main/application/pdes_compare/README.md

Past and Ongoing Research Use

SerializableSimpy was developed as part of a research collaboration between the University of Luxembourg and the Goodyear Company. It is currently being integrated into workflows for decomposed, large-scale discrete-event manufacturing simulations, enabling execution through parallel logical processes on multi-core or networked environments. Its simple Python API and minimal software dependencies facilitate rapid prototyping and experimentation. These lightweight dependencies are compatible with Python Just-In-Time (JIT) compilers and have been tested with (Bolz et al., 2009), enhancing performance for many multi-core and distributed workloads.

While SerializableSimpy provides examples and tools to facilitate parallel execution with minimal user effort, users are still required to manually decompose simulations and assign tasks to processing cores. To reduce this burden, ongoing research explores graph-based decomposition techniques (Karypis & Kumar, 1998), with the goal of automating partitioning and enhancing scalability, evaluated on large-scale manufacturing simulations.

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